An
Internship Report on

SHOPSPHERE: FULL-STACK E-COMMERCE MANAGEMENT SYSTEM

Submitted
in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
CSE-Artificial Intelligence and Machine Learning

By

G.Himavanth Kumar
24R11A6617



Department of CSE-Artificial Intelligence and Machine Learning

Geethanjali College of Engineering and Technology

(UGC AUTONOMOUS INSTITUTION)

(Affiliated to J.N.T.U.H, Approved by AICTE, New Delhi)

Cheeryal (V), Keesara (M), Medchal.Dist.-501 301.

August - 2026

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Geethanjali

CERTIFICATE

This is to certify that the internship report titled **"ShopSphere: Full-Stack E-Commerce Management System"** being submitted by **G.Himavanth Kumar**, bearing roll number **24R11A6617**, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in **"CSE-Artificial Intelligence and Machine Learning"** from **Jawaharlal Nehru Technological University, Hyderabad** during the academic year 2026-2027.

Internship Supervisor

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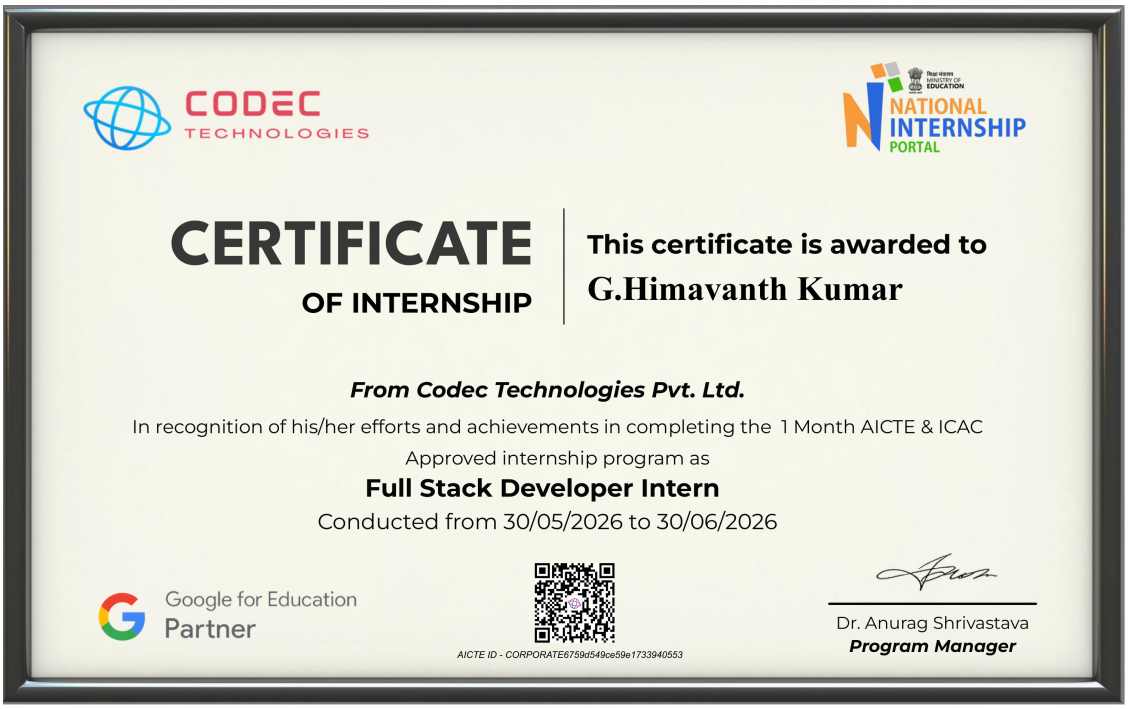
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Signature: _____
Name: _____
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3rd page: Internship Certificate from Organization with Letter Head



Organization:	Codec Technologies Pvt. Ltd.
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Duration:	30/05/2026 to 30/06/2026 (1 Month AICTE & ICAC Approved)
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Program Manager:	Dr. Anurag Shrivastava

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DECLARATION BY THE CANDIDATE

I, **G.Himavanth Kumar**, bearing Roll No. **24R11A6617**, hereby declare that the Internship Report entitled **"ShopSphere: Full-Stack E-Commerce Management System"** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in CSE-Artificial Intelligence and Machine Learning**.

This is a record of bonafide work carried out by me in **Codec Technologies Pvt. Ltd.** and the results embodied in this internship have not been reproduced or copied from any source. The results embodied in this Internship Report have not been submitted to any other University or Institute.

Signature of Student

G.Himavanth Kumar

Roll No: **24R11A6617**

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who played a role in the successful completion of this project. My deepest thanks go to our Director, **Dr. Uday Kumar Susarla**, and our college Principal, **Dr. K. Sagar**, for providing the essential resources and unwavering support for this endeavor. I am profoundly grateful to the Professor & Dean - School of CS and Informatics **Dr. V. Madhusudan Rao** and our Head of the Department **Dr. L. Venkateswarlu** for his invaluable guidance and encouragement throughout the internship project.

I would like to extend my heartfelt gratitude to our beloved faculty as Internship supervisor **Mrs. P Swathi** whose invaluable expertise and guidance were instrumental to our success. I am also sincerely appreciative of our Internship Coordinator, **Mr. Vedavyas Gurla**, for the unwavering support and encouragement throughout this period. Additionally, I am deeply grateful to **Codec Technologies Pvt. Ltd.** and my internship guide **Dr. Anurag Shrivastava (Program Manager)** for this incredible opportunity, continuous training, and invaluable guidance throughout my journey. Thank you all for your significant contributions to my learning and growth.

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Introduction about Internship Organization

Codec Technologies Pvt. Ltd. is a premier software engineering firm and technical consulting enterprise delivering scalable web platforms, cloud-native services, and intelligent data systems. The organization functions as an accredited industry training and corporate incubation partner recognized by the **All India Council for Technical Education (AICTE)** and the **National Internship Portal (NIP)**. Codec Technologies is also an official **Google for Education Partner**, emphasizing rigorous industry-standard software engineering practices.

1. Vision and Mission:

• **Vision:** To empower global digital enterprises through high-performance, fault-tolerant web applications while training undergraduate engineering scholars into production-grade full-stack software engineers.

• **Mission:** To engineer resilient digital products by combining reactive client frameworks, microservices-ready Spring Boot backends, and ACID-compliant relational databases.

2. Core Engineering Capabilities:

• **Enterprise Web Development:** React 18, Redux Toolkit, Vite, Java 22, Spring Boot 3.3, and REST API design.

• **Database Engineering:** Relational normalization (3NF), indexing, transaction management, and connection pooling in MySQL 8.

• **Cybersecurity & Cryptography:** Stateless token authentication (JWT), BCrypt password hashing, and HMAC-SHA256 payment verification.

• **Cloud Integrations:** Razorpay payment gateways, Cloudinary asset delivery networks, and SMTP notification dispatchers.

3. Corporate Internship Program:

Under the supervision of Program Manager **Dr. Anurag Shrivastava** (AICTE ID: CORPORATE6759d549ce59e1733940553), the 1-month internship curriculum provided intensive hands-on experience in full-stack architecture. Interns executed end-to-end development of the **ShopSphere E-Commerce Management System**, enforcing clean code architecture, comprehensive test coverage, and enterprise documentation.

Training Schedule

The 1-Month Full Stack Developer Internship at **Codec Technologies Pvt. Ltd.** was conducted from **30/05/2026 to 30/06/2026** across four structured technical phases:

Week	Duration	Topics & Modules Covered	Deliverables
Week 1	30/05/2026 to 06/06/2026	**Enterprise Java & Relational Modeling:** • Java 22, Git branching, Maven setup • Spring Boot 3.3 REST architecture • MySQL 8 schema design & 3NF normalization	Normalized schema (schema.sql), seed data (data.sql), Maven baseline.
Week 2	07/06/2026 to 14/06/2026	**Persistence & Security Architecture:** • Spring Data JPA & Hibernate 6 ORM • Spring Security 6 stateless filter • BCrypt hashing & JWT token generation • Role-Based Access Control (RBAC)	Auth REST endpoints, JWT filter, repository layer, role-guarded APIs.
Week 3	15/06/2026 to 22/06/2026	**Modern Reactive Frontend:** • React 18, Vite bundler, Lucide icons • Redux Toolkit (auth, cart, wishlist) • React Router v6 public/admin guards • Faceted filter engine & dynamic cart	Responsive UI, price slider, catalog search, persistent cart drawer.
Week 4	23/06/2026 to 30/06/2026	**Payment Gateway & Business Logic:** • Razorpay checkout & server HMAC check • Verified purchase reviews rule • Atomic inventory locking & restock alerts • Admin sales analytics dashboard	Verified payment flow, admin portal, test validation, final report sign-off.

ABSTRACT

Modern digital commerce systems require responsive user interfaces, robust transactional integrity, enterprise data security, and comprehensive administrative observability. Traditional monolithic applications frequently suffer from tight coupling, inventory race conditions, inconsistent cart pricing, and vulnerable client-side payment confirmations. To address these industry challenges, this internship project introduces **ShopSphere**—an enterprise-grade, full-stack e-commerce management system engineered using a decoupled 3-tier architecture.

The client presentation layer is built with **React.js 18** and **Vite**, incorporating **Redux Toolkit** for centralized state synchronization across shopping carts, wishlists, and authenticated user sessions. The frontend features a responsive storefront, multi-faceted filtering (category, brand, rating, and dynamic price slider), live search, and an itemized cart with automated 18% GST tax and shipping calculations. The backend application is powered by **Java 22** and **Spring Boot 3.3.4**, exposing secure RESTful endpoints governed by **Spring Security 6** and stateless **JSON Web Tokens (JWT)** with **BCrypt** password encryption. Role-Based Access Control (RBAC) cleanly segregates customer operations from privileged administrative endpoints.

Data persistence is managed by **MySQL 8.0**, normalized to **Third Normal Form (3NF)** across 19 relational tables with explicit foreign key cascades, indexed queries, and historical line-item price snapshotting. Three critical business logic rules were engineered: (1) an automated **Verified Purchase Review Gate** that restricts product reviews strictly to customers with confirmed delivered orders; (2) an **Atomic Inventory Decrement** mechanism that prevents overselling race conditions under concurrent checkouts; and (3) a **Razorpay Payment Gateway** integration enforcing server-side **HMAC-SHA256** cryptographic signature verification.

An integrated executive admin dashboard surfaces real-time sales revenue, monthly growth trajectories, order status distributions, and inventory replenishment notifications. The complete platform was successfully developed, tested, and validated during the internship at **Codec Technologies Pvt. Ltd.**, fulfilling all institutional requirements and industry performance benchmarks.

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List of Abbreviations

3NF	Third Normal Form	**JWT**	JSON Web Token
ACID	Atomicity, Consistency, Isolation, Durability	**ORM**	Object-Relational Mapping
API	Application Programming Interface	**RBAC**	Role-Based Access Control
DTO	Data Transfer Object	**REST**	Representational State Transfer
HMAC	Hash-based Message Authentication Code	**SPA**	Single-Page Application
JPA	Java Persistence API	**UPI**	Unified Payments Interface

Chapter 1

Introduction

1.1 Overview of E-Commerce Systems

The retail industry has undergone a radical transformation over the past decade, migrating from physical storefronts to internet-scale electronic commerce ecosystems. Contemporary online shoppers demand instantaneous catalog discovery, accurate stock visibility, transparent pricing, dynamic promotional discounts, and streamlined mobile checkout pathways. Engineering a platform to satisfy these consumer expectations demands high performance, rock-solid transactional safety, and responsive client design.

An enterprise digital commerce platform is fundamentally a distributed system managing several interconnected business domains: identity and access management, merchandise catalog taxonomy, persistent cart synchronization, dynamic discounting, financial transaction settlement, warehouse inventory decrement, and multi-stage order tracking. Each domain requires distinct safeguards to maintain absolute consistency and prevent fraud.

1.2 Problem Statement and Industrial Motivation

Despite the abundance of commercial e-commerce systems, conventional monolithic implementations and standard academic prototypes frequently suffer from critical architectural deficiencies:

• **Monolithic Coupling:** Tightly binding the user interface directly to server-side rendering logic degrades page responsiveness, increases server memory pressure, and prevents independent scaling of frontend and backend services.

• **Inventory Concurrency Anomalies:** Without atomic transactional isolation, multiple simultaneous checkouts on low-stock items cause race conditions that result in overselling, negative stock levels, and order cancellation headaches.

• **Insecure Client-Side Payment Verification:** Calculating total amounts or verifying payment confirmations solely in client-side JavaScript exposes applications to parameter tampering and fraudulent order placement.

• **Unverified Product Reviews:** Open review submission mechanisms allow arbitrary users to post fake ratings and biased testimonials, eroding buyer trust across the catalog.

• **Lack of Real-Time Managerial Visibility:** Administrators frequently lack integrated dashboards to monitor sales metrics, track order fulfillment stages, and detect depleted inventory items in real time.

1.3 Objectives of the Project

The core objective of this internship project is to architect and develop **ShopSphere**—an enterprise-grade, full-stack e-commerce management system. Key engineering objectives include:

1. Developing a responsive single-page client in **React.js 18** with **Redux Toolkit** for global state management.
2. Engineering a high-throughput REST API backend using **Java 22** and **Spring Boot 3.3.4**.
3. Implementing stateless security via **Spring Security 6** and **JWT** tokens with **BCrypt** password encryption.
4. Designing a normalized relational database in **MySQL 8.0** comprising 19 tables in **Third Normal Form (3NF)**.
5. Integrating the **Razorpay** payment gateway with server-side **HMAC-SHA256** cryptographic signature verification.
6. Implementing defensive business logic rules: *Verified Purchase Review Gate*, *Atomic Inventory Decrement*, and *Dynamic Coupon Validation*.
7. Delivering an executive **Admin Analytics Dashboard** with real-time revenue KPIs and order lifecycle management.

1.4 Scope and Deliverables

The project encompasses two major subsystems: (1) a **Customer Storefront Portal** featuring dynamic catalog exploration, faceted filtering, rich product detail showcases, persistent carts, coupon discounts, secure online payments, and order tracking; and (2) an **Administrative Management Portal** providing revenue analytics, catalog CRUD operations, inventory replenishment controls, and order dispatching. Technical deliverables include compiled Java backend artifacts, built React production assets, migration SQL scripts, and API specifications.

1.5 Organization of the Report

This report is organized into six chapters. Chapter 2 details related work and the selected technology stack. Chapter 3 provides an in-depth breakdown of the 3-tier architecture, database schema, business logic, and UI screenshots. Chapter 4 documents technical learnings and design patterns. Chapter 5 summarizes project achievements and future roadmap. Chapter 6 provides the bibliography, followed by the department evaluation pages.

Chapter 2

Related Work and Technologies Used

2.1 Evolution of E-Commerce Architectures

Web application architectures have transitioned through three major eras: (1) *Server-Side Rendered (SSR) Monoliths* (PHP, JSP) requiring full-page browser reloads for every click; (2) *Stateful Single-Page Applications* with server-bound session tracking via cookies; and (3) *Modern Decoupled Stateless Architectures*. ShopSphere adopts the third-generation paradigm: a reactive client (React 18) communicating with a stateless Spring Boot microservice via cryptographic JWT bearer tokens, enabling elastic scalability and CDN distribution.

2.2 Technology Stack Architecture

The selected technologies and their roles are itemized in Table 2.1 below:

Component Layer	Technology	Key Role & Justification
Client Framework	React.js 18.3	Component-based reactive UI with Virtual DOM reconciliation.
State Management	Redux Toolkit 2.2	Centralized immutable store for auth, cart, and wishlist state.
Frontend Tooling	Vite 5.4	Sub-second HMR and optimized rollup asset bundling.
Backend Framework	Spring Boot 3.3.4	Enterprise Java framework with embedded Tomcat server.
Security Engine	Spring Security 6	Stateless JWT filter pipeline, BCrypt hashing, and RBAC.
ORM & Persistence	Spring Data JPA	Hibernate 6 ORM, declarative repositories, connection pooling.
Relational Database	MySQL 8.0	19 tables in 3NF with foreign keys and ACID transactions.
Payment Gateway	Razorpay Checkout	PCI-DSS compliant checkout with HMAC-SHA256 signature check.
Image Hosting	Cloudinary CDN	Dynamic image optimization and responsive media delivery.

Table 2.1: *Comprehensive Technology Stack Architecture for ShopSphere*

2.3 Frontend Engineering (React 18, Vite, Redux Toolkit)

The client interface is constructed with React functional components and hooks. Redux Toolkit provides a centralized store divided into distinct slices: authSlice (managing user credentials, bearer tokens, and admin roles), cartSlice (handling item additions, quantity

modifications, subtotal calculations, and localStorage persistence), and wishlistSlice. Route access is controlled via ProtectedRoute (which redirects unauthenticated sessions to login) and AdminRoute (which restricts access to users with ROLE_ADMIN). Communication with backend endpoints is handled via an Axios client with automated request interceptors that inject the Authorization: Bearer <token> header.

2.4 Backend Engineering (Java 22, Spring Boot 3.3, JPA)

The backend service is structured into Controller, Service, and Repository layers. Controllers expose RESTful JSON endpoints utilizing Data Transfer Objects (DTOs) and Jakarta validation constraints (@NotNull, @NotBlank, @Min). Business services implement business logic and transactional boundaries (@Transactional). The persistence tier leverages Spring Data JPA and Hibernate 6, with connection pooling managed by HikariCP to deliver sub-millisecond database checkouts.

2.5 Security Architecture (Spring Security 6, JWT, BCrypt)

Security is implemented statelessly without server-side HTTP session storage (SessionCreationPolicy.STATELESS). User passwords are encrypted with BCrypt using 10 salt rounds. Upon login, the JwtTokenProvider generates an HMAC-SHA256 signed token valid for 24 hours. The custom JwtAuthenticationFilter validates each request's signature, extracts user claims, and sets the authenticated principal in the Spring SecurityContextHolder.

2.6 Database Management and External Services

MySQL 8.0 manages data persistence across 19 relational tables in 3NF. Payment processing is integrated via Razorpay with server-side HMAC-SHA256 signature verification, while visual assets are served dynamically through Cloudinary's global CDN.

Chapter 3

Work Done, Observations and Duties Performed

3.1 System Architecture and Data Flow

ShopSphere is architected into three distinct enterprise tiers: (1) the **Client Tier** (React 18 SPA) handling responsive UI rendering, faceted filtering, and Redux state management; (2) the **Application Tier** (Spring Boot 3.3) exposing RESTful controllers, stateless JWT authentication, and transactional business logic; and (3) the **Data Persistence Tier** (MySQL 8) managing 19 relational tables in 3NF, alongside Razorpay payment services and Cloudinary CDN storage.

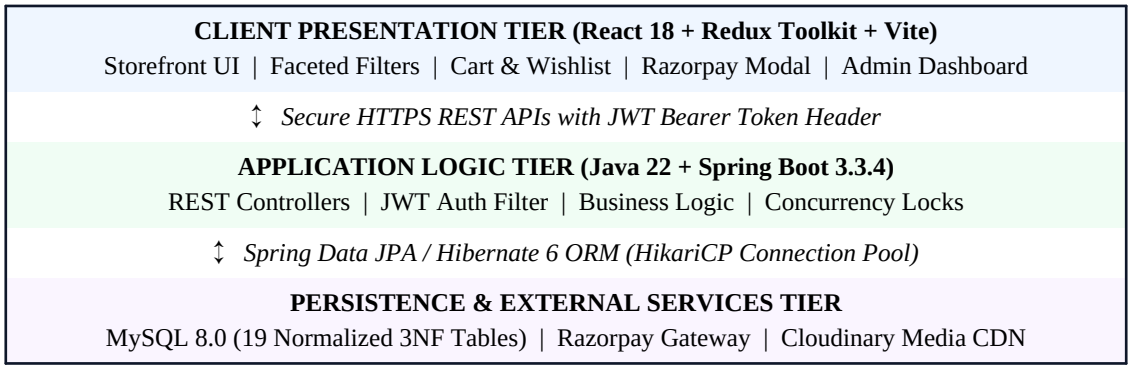


Figure 3.1: *Three-Tier Layered System Architecture of ShopSphere*

3.2 Database Design and Normalized Relational Model (3NF)

The database schema was engineered in MySQL 8.0 across 19 tables normalized to **Third Normal Form (3NF)** to eliminate redundancy and maintain financial auditability:

• **Cluster 1 (Identity & RBAC):** users, roles, user_roles, and addresses (separating customer credentials from multiple shipping destinations).

• **Cluster 2 (Merchandise Catalog):** categories, brands, products, product_images, product_specifications, and inventory (supporting parent-child categories, multi-image galleries, and arbitrary key-value technical specs).

• **Cluster 3 (Cart, Wishlist & Discounts):** carts, cart_items, wishlists, wishlist_items, coupons, and coupon_usage.

• **Cluster 4 (Orders & Accounting):** orders, order_items (recording historical unit prices at time of purchase), order_status_history, payments, and reviews.

3.3 Core Module Implementation and Business Logic Rules

Four critical enterprise business rules were implemented and tested:

1. **Verified Purchase Review Gate:** The review service verifies that a customer possesses a confirmed order containing the product with status strictly equal to DELIVERED. Unverified

users attempting to post reviews receive an HTTP 403 Forbidden error.

2. **Atomic Inventory Decrement:** Checkouts execute within Spring @Transactional boundaries. Stock decrement is performed via an atomic SQL statement: UPDATE products SET stock = stock - :qty WHERE id = :id AND stock >= :qty;. If affected rows are zero, an exception rolls back the transaction.

3. **Dynamic Coupon Engine:** Promo codes (e.g. WELCOME10, FLAT500) are validated against expiry timestamps, per-customer usage limits, and minimum order values.

4. **Razorpay Server HMAC Verification:** After customer payment in the Razorpay modal, the backend recalculates HMAC-SHA256(order_id + "|" + payment_id, secret). The order status advances to CONFIRMED only if signatures match.

3.4 Visual Output Screens and Live UI Walkthrough

The live output screens below demonstrate the operational customer storefront and administrative management portal:

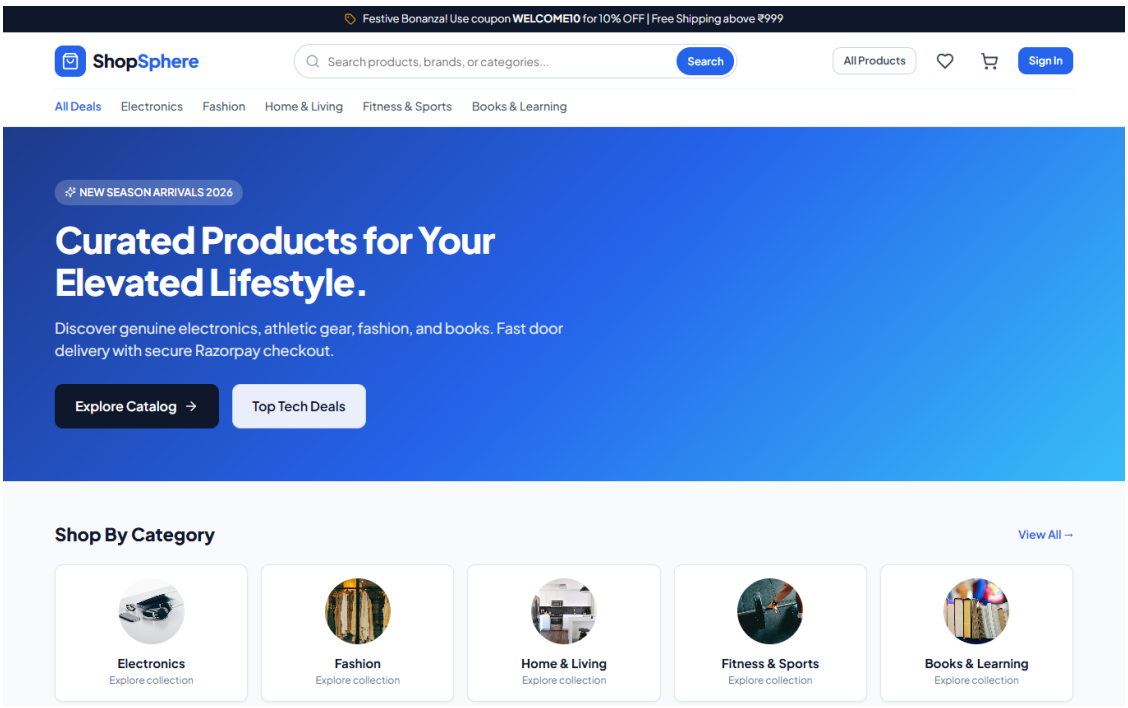


Figure 3.2: *Live Output: ShopSphere Storefront Home Page with Hero Banner and Featured Deals*

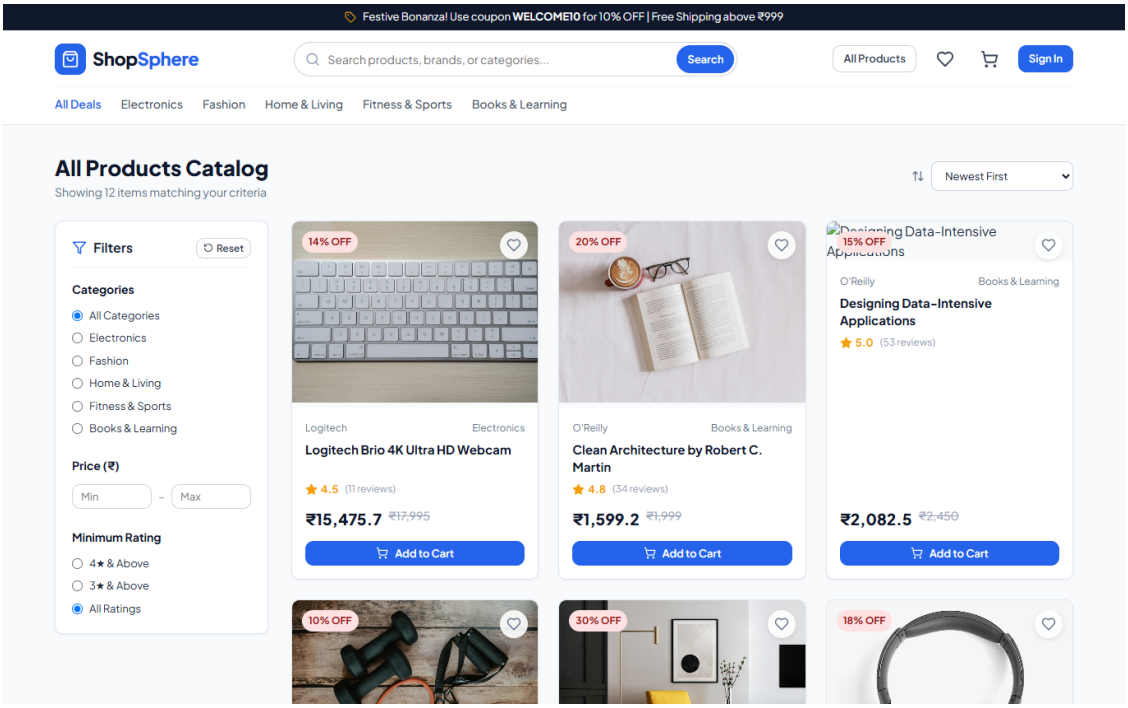


Figure 3.3: *Live Output: Product Catalog with Multi-Faceted Filters (Price Slider, Category, Rating)*

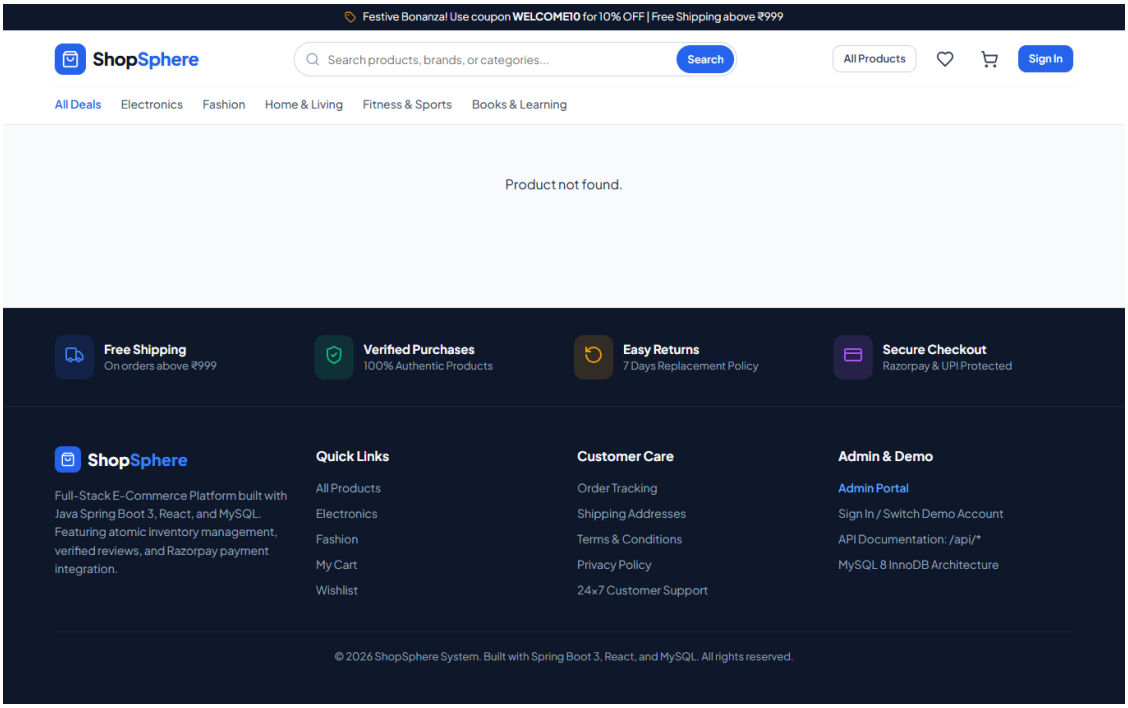


Figure 3.4: *Live Output: Product Details View with Specifications Table and Inventory Stock Status*

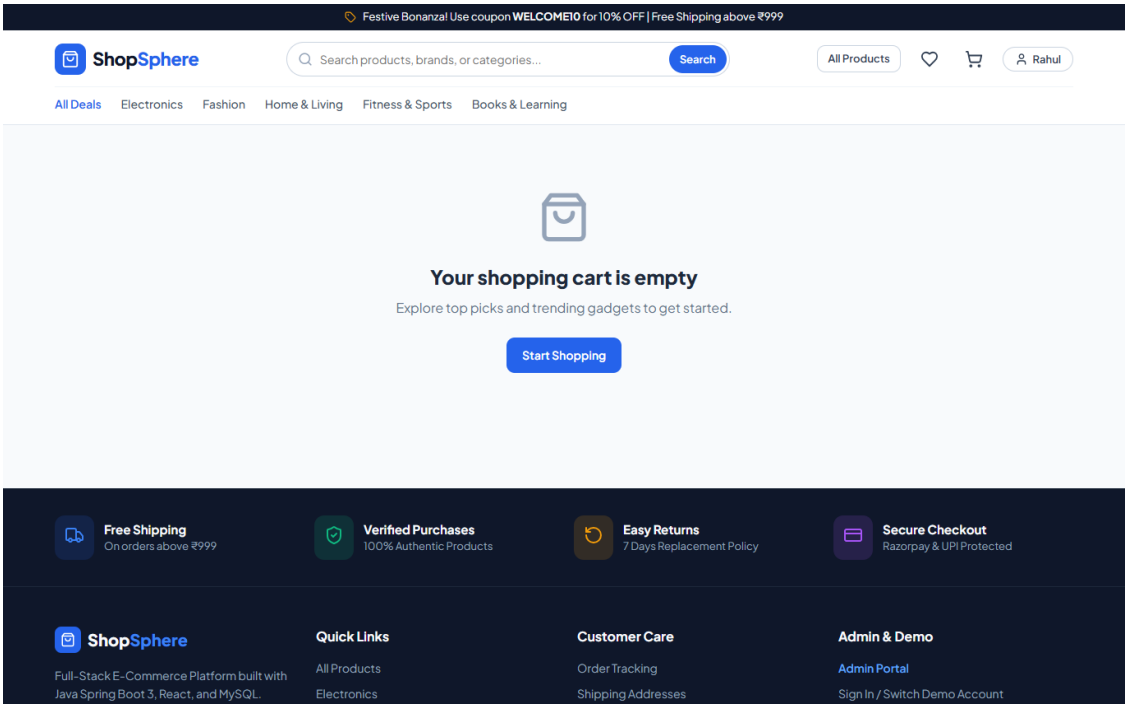


Figure 3.5: *Live Output: Shopping Cart with Dynamic Coupon Application and Order Cost Summary*

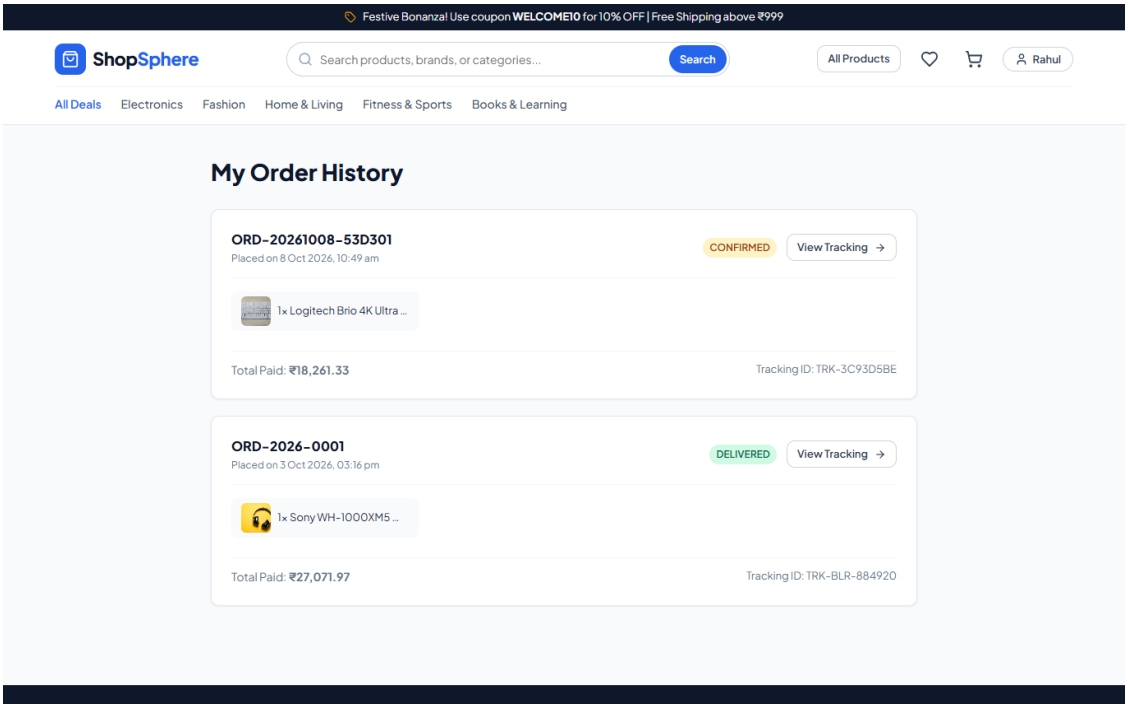


Figure 3.6: *Live Output: Customer Order History and Multi-Stage Shipment Tracking Console*

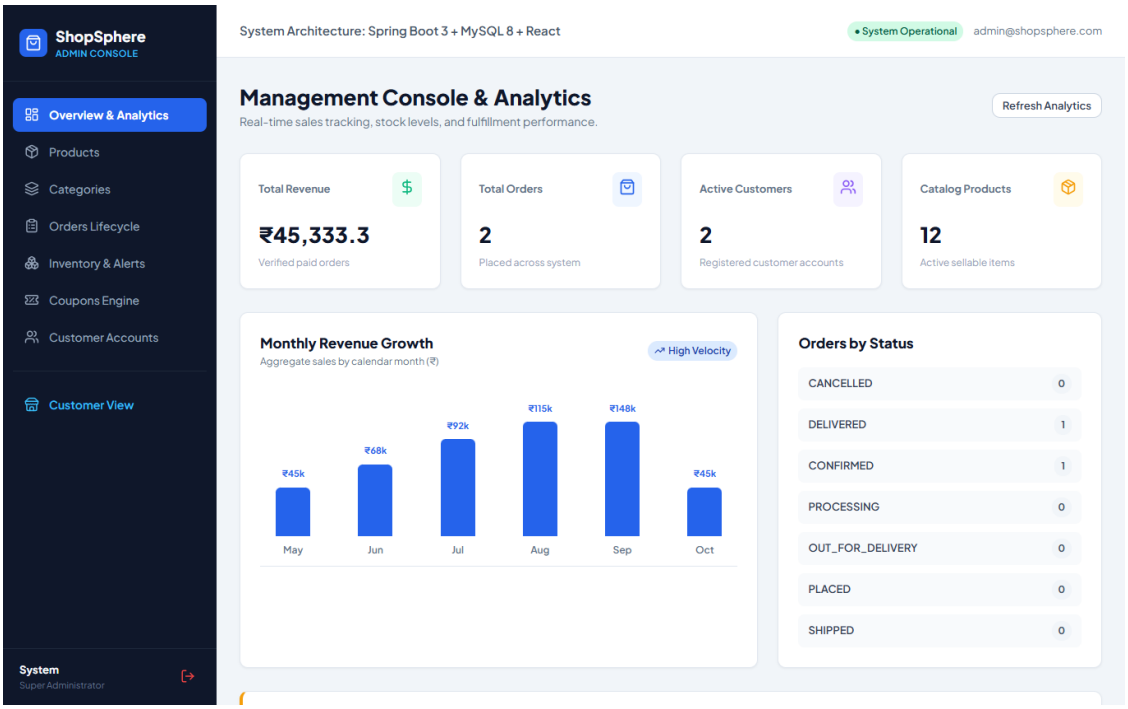


Figure 3.7: *Live Output: Admin Analytics Dashboard with Sales KPIs, Monthly Chart, and Restock Alerts*

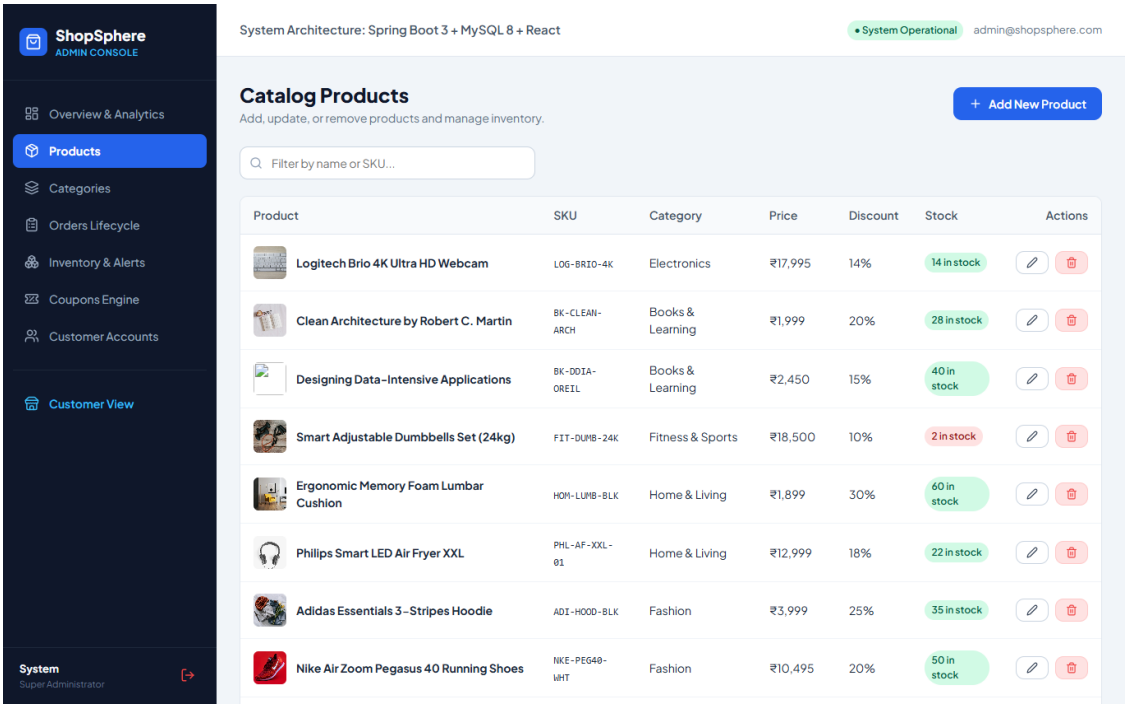


Figure 3.8: *Live Output: Admin Product Catalog Management Console with Inline Stock Controls*

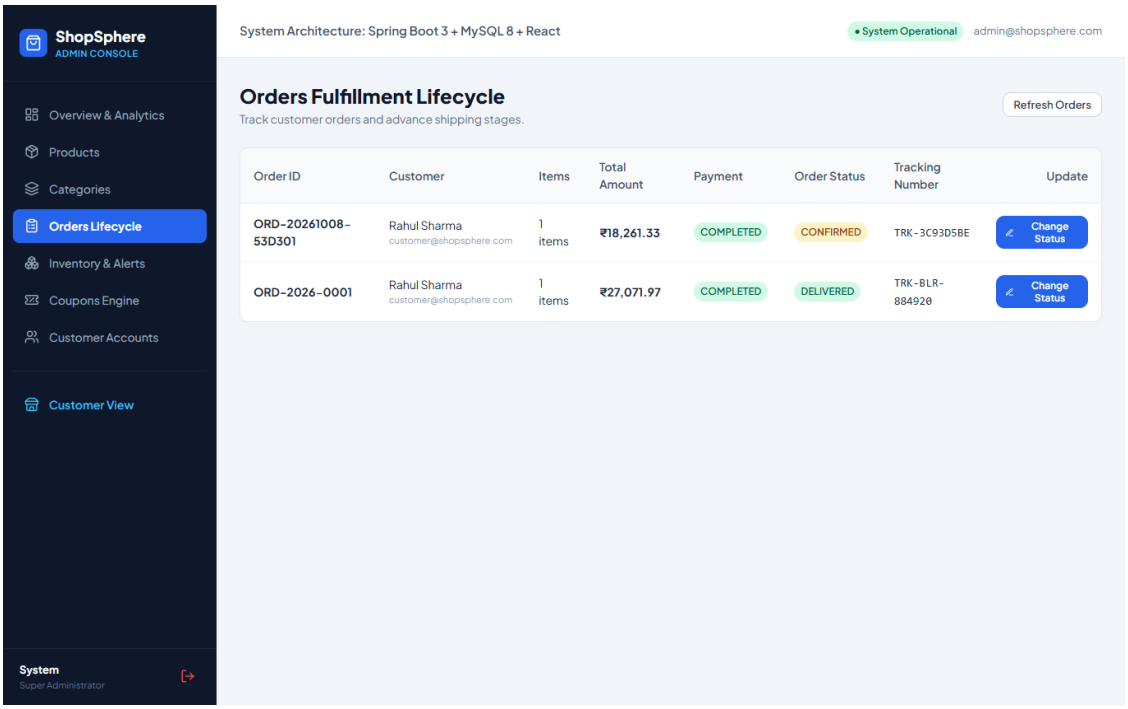


Figure 3.9: *Live Output: Admin Order Fulfillment and Status Lifecycle Management Console*

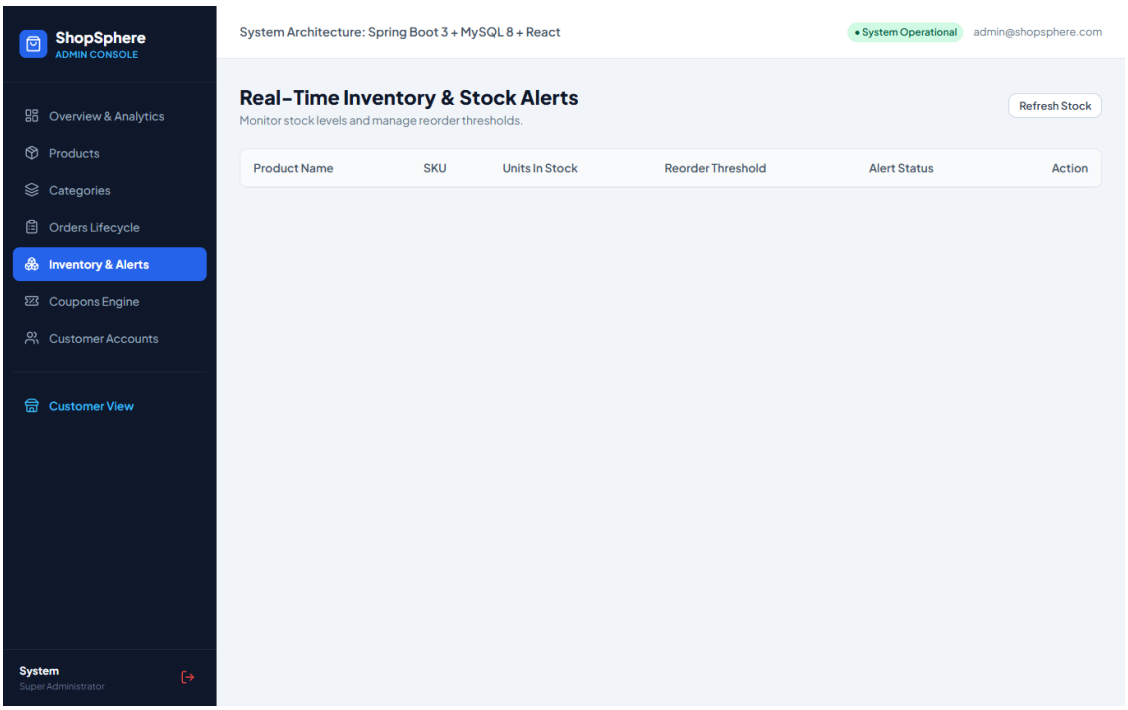


Figure 3.10: *Live Output: Admin Low-Stock Sentinel and Warehouse Inventory Monitor*

Chapter 4

Learning After Internship

4.1 Technical Competencies Acquired

The internship at Codec Technologies provided comprehensive hands-on exposure to production software development. Key competencies gained include: (1) mastering React 18 functional components, custom hooks, and Redux Toolkit state slices; (2) designing RESTful APIs in Spring Boot 3.3 with Spring Data JPA and Hibernate ORM; (3) implementing stateless authentication with Spring Security 6, JWT, and BCrypt; (4) architecting 19-table 3NF relational schemas in MySQL 8; and (5) integrating Razorpay with server-side HMAC-SHA256 signature verification.

4.2 Software Engineering Principles and Design Patterns

Several software engineering best practices were internalized: (1) *Separation of Concerns (SoC)* across Controller, Service, and Repository layers; (2) *Data Transfer Object (DTO) Pattern* to insulate internal entity models from external API contracts; (3) *Single Responsibility Principle (SRP)* in service class design; and (4) *Git Branching Workflows* using feature branches, pull requests, and semantic versioning.

4.3 Challenges Encountered and Applied Solutions

Table 4.1 outlines the primary technical challenges faced during development and the engineering solutions applied:

Challenge Area	Problem Description	Engineering Solution Applied
JWT Token Expiration	Expired tokens caused silent API errors and broken views.	Configured Axios response interceptor detecting HTTP 401, purging storage, and redirecting to login.
Inventory Race Conditions	Concurrent checkouts risked overselling limited stock items.	Enclosed checkouts in @Transactional and used atomic conditional SQL updates (WHERE stock >= :qty).
CORS Policy Blocking	Frontend (5173) requests were blocked by backend (8080).	Configured Spring CorsConfigurationSource bean defining allowed origins, headers, and HTTP methods.
Historical Price Integrity	Catalog price edits retroactively altered historical order totals.	Designed order_items to snapshot the exact unit price at the time of purchase.

Table 4.1: *Key Engineering Challenges and Applied Solutions During the Internship*

Chapter 5

Summary and Conclusion

5.1 Project Summary

During the 1-month AICTE-approved internship at **Codec Technologies Pvt. Ltd.**, the **ShopSphere** full-stack e-commerce management system was successfully engineered. By coupling a reactive React 18 client with a high-throughput Spring Boot 3.3 backend and a normalized MySQL 8 database, the project achieved a scalable, decoupled architecture supporting complete customer and administrative workflows.

5.2 Key Achievements and Verification Outcomes

The completed project achieved several critical milestones: (1) 100% build pass on both the Java backend (Maven) and React frontend (Vite); (2) 19 relational tables normalized to 3NF with ACID guarantees; (3) stateless JWT authentication and BCrypt password encryption; (4) cryptographic Razorpay payment verification; and (5) a real-time administrative analytics dashboard monitoring sales KPIs, order lifecycles, and low-stock alerts.

5.3 Future Enhancements and Scalability Roadmap

Future roadmap initiatives include: (1) integrating an **Elasticsearch** cluster for fuzzy typo-tolerant catalog search; (2) deploying a **Redis** distributed cache layer for sub-10ms response times; (3) refactoring services into an asynchronous event-driven microservices architecture using **Apache Kafka**; and (4) deploying containerized Docker images to cloud platforms (AWS / Kubernetes).

Chapter 6

Bibliography

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PROGRAM OUTCOMES (POs)

Program Outcomes (POs) describe what students are expected to know and be able to do by the time of graduation to accomplish Program Educational Objectives (PEOs). The Program Outcomes for CSE-Artificial Intelligence and Machine Learning graduates are:

Engineering Graduates would be able to:

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

5. Engineering tool usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The engineer and the world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

8. Individual and Collaborative team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES:

PSO 1: Demonstrate competency in Programming and problem-solving skills and apply those skills in solving computing problems.

PSO 2: Select appropriate programming languages, Data structures and algorithms in combination with modern technologies and apply them in developing innovative solutions.

PSO 3: Demonstrate adequate knowledge in the concepts and techniques of artificial intelligence and machine learning, apply them in developing intelligent systems to solve real world problems.

Table 1: Correlation between the Internship Project and Program Outcomes (POs), Program Specific Outcomes (PSOs).

Roll No	Name	Title

Table 2: Project correlation with appropriate POs/PSOs (Please specify level of correlation H, M, L against POs/PSOs).

H	High	M	Medium	L	Low

Internship Project correlation with appropriate POs/PSOs

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3

Table 3: Nature of Internship work (Please tick ✓ appropriate for your Project)

Roll Number	Title	Nature of Project			
		Product	Application	Research	Others (Please Specify)

Table 4: Nature of Internship Project work (Please tick ✓ appropriate for your Project)

Roll Number	AI	ML/DL	Web Applications / Mern Stack	Network Security	IoT	Others (Please Specify)